

DIGITAL RADIO LABORATORY

Implementation of OOK, ASK, and PSK Digital Modulation and Demodulation using GNU Radio and ADALM-Pluto SDR

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1. Aim

To implement On-Off Keying (OOK), Amplitude Shift Keying (ASK), and Phase Shift Keying (PSK) digital modulation and demodulation schemes using GNU Radio, and to demonstrate real-time RF transmission and reception using the ADALM-Pluto Software Defined Radio (SDR). To analyse the transmitted and received signals in both the time and frequency domains, and to compare the performance of the three modulation schemes on the basis of their constellation diagrams.

2. Software Used

- GNU Radio Companion (GRC) – for designing the transmitter and receiver flow-graphs using drag-and-drop signal processing blocks.
- ADALM-Pluto SDR – used as the RF front end for actual over-the-air transmission and reception.
- Pluto SDR Source / Sink blocks (gr-iio) – GNU Radio blocks interfacing with the ADALM-Pluto hardware.
- QT GUI Time Sink, Frequency Sink and Constellation Sink – for real time visualisation of the signals.
- Operating System: Ubuntu Linux (GNU Radio runtime environment).

3. Theory

3.1 Digital Modulation

Digital modulation is the process of mapping digital information (bits) onto the amplitude, frequency, or phase of a high-frequency carrier signal so that it can be transmitted efficiently over a communication channel or through free space using an antenna. In this experiment, three simple and widely used binary modulation schemes — OOK, ASK, and PSK — are implemented and studied using a Software Defined Radio platform.

3.2 On-Off Keying (OOK)

On-Off Keying is the simplest form of amplitude modulation in which a binary '1' is represented by the presence of the carrier signal (full amplitude) and a binary '0' is represented by the complete absence of the carrier (zero amplitude). Mathematically, it can be viewed as a special case of ASK where the amplitude for logic '0' is zero. OOK is easy to generate and detect (using a simple envelope detector), but it is more susceptible to noise since the '0' symbol carries no energy, making it harder to distinguish from a faded or noisy channel.

3.3 Amplitude Shift Keying (ASK)

Amplitude Shift Keying represents digital data by varying the amplitude of the carrier wave between two (or more) discrete non-zero levels, corresponding to the binary symbols. Unlike OOK, both logic levels carry some carrier energy, which generally improves noise immunity slightly. ASK is generated by multiplying the baseband bit stream (mapped to the desired amplitude levels) with a carrier signal, and is demodulated using envelope detection or coherent detection followed by a decision (threshold) device.

3.4 Phase Shift Keying (PSK)

Phase Shift Keying conveys digital information by changing the phase of the carrier signal while keeping its amplitude constant. In Binary PSK (BPSK), a logic '1' and a logic '0' are represented by two carrier phases that are 180° apart (e.g., 0° and 180°). Because the signal amplitude remains constant, PSK is more robust to amplitude-based noise and channel fading compared to OOK/ASK, and it generally provides better bit error rate performance for the same signal-to-noise ratio. Demodulation of PSK requires a coherent receiver capable of tracking the carrier phase.

3.5 GNU Radio

GNU Radio is an open-source software development toolkit that provides signal processing blocks to implement software radios. Using GNU Radio Companion (GRC), a graphical tool, complete transmitter and receiver chains can be built by dragging, dropping, and connecting pre-built blocks (source, modulator, filter, SDR sink/source, demodulator, GUI sink, etc.) without writing manual code. The resulting flow-graph is automatically translated into an executable Python/C++ signal processing chain.

3.6 ADALM-Pluto SDR

The ADALM-Pluto (PlutoSDR) is a low-cost, USB-powered Software Defined Radio learning tool built around the AD9363 RF transceiver. It provides a single transmit and a single receive RF channel covering a wide tunable frequency range, and interfaces with GNU Radio through the gr-iio blocks (Pluto SDR Sink for transmission and Pluto SDR Source for reception). It allows real over-the-air RF transmission and reception, which enables practical, hardware-in-the-loop verification of the modulation and demodulation schemes designed in GNU Radio.

3.7 Key Processing Blocks Used

- Random/ Vector Source – generates the baseband bit stream to be transmitted.
- Modulator (Chunks-to-Symbols / Multiply) – maps bits to the required amplitude/phase levels for OOK, ASK or PSK.
- Pluto SDR Sink – transmits the modulated complex baseband signal over the air.
- Pluto SDR Source – receives the RF signal and down-converts it to complex baseband.
- AGC (Automatic Gain Control) – normalises the amplitude of the received signal before demodulation.
- Constellation Decoder – maps the received complex samples back to the nearest ideal constellation point / bit decision.
- Keep 1 in N – down-samples the signal stream, generally used for display or rate reduction.
- QT GUI Time/Frequency/Constellation Sinks – used to visualise the signal in time domain, frequency domain and as an I-Q constellation.

4. Source Code

This experiment was implemented entirely using GNU Radio Companion (GRC), a graphical, block-based design environment. The transmitter and receiver chains for OOK, ASK, and PSK were built by dragging, dropping and interconnecting the required signal-processing and Pluto SDR blocks — no manual source code was written by hand. The resulting GRC block diagrams (flow-graphs) for each of the three modulation schemes are given below.

4.1 OOK — GNU Radio Companion Block Diagram

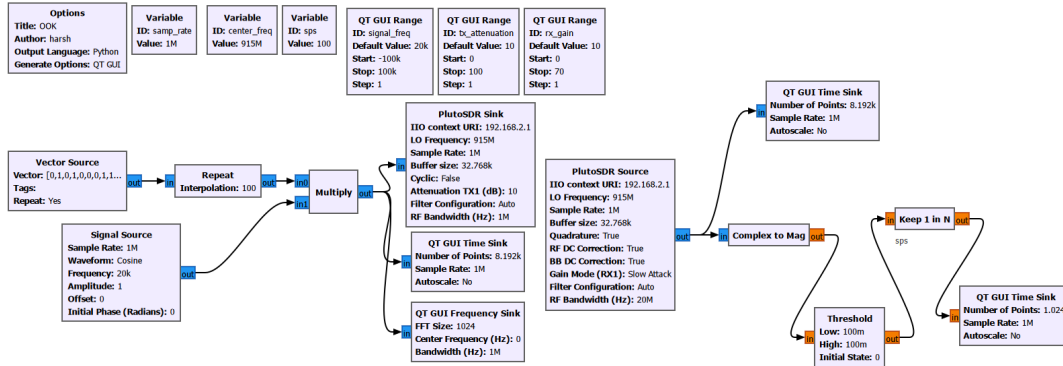


Fig. B2: GNU Radio Companion flow-graph for OOK transmitter and receiver.

4.2 ASK — GNU Radio Companion Block Diagram

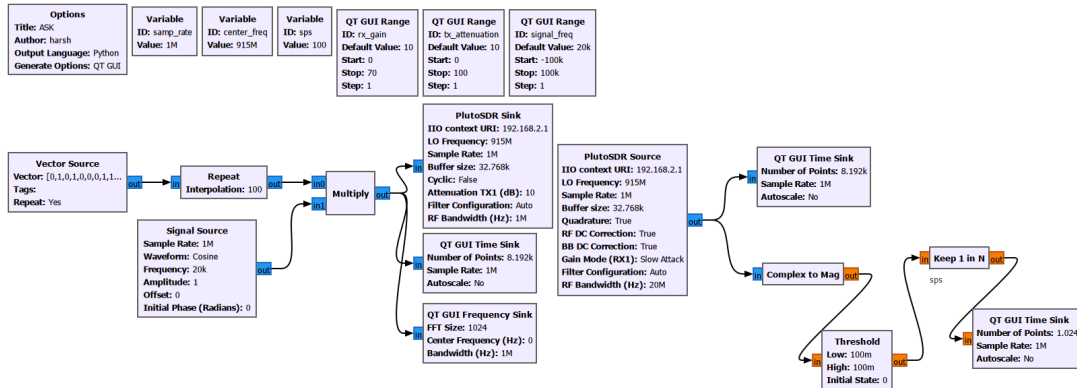


Fig. B1: GNU Radio Companion flow-graph for ASK transmitter and receiver.

4.3 PSK — GNU Radio Companion Block Diagram

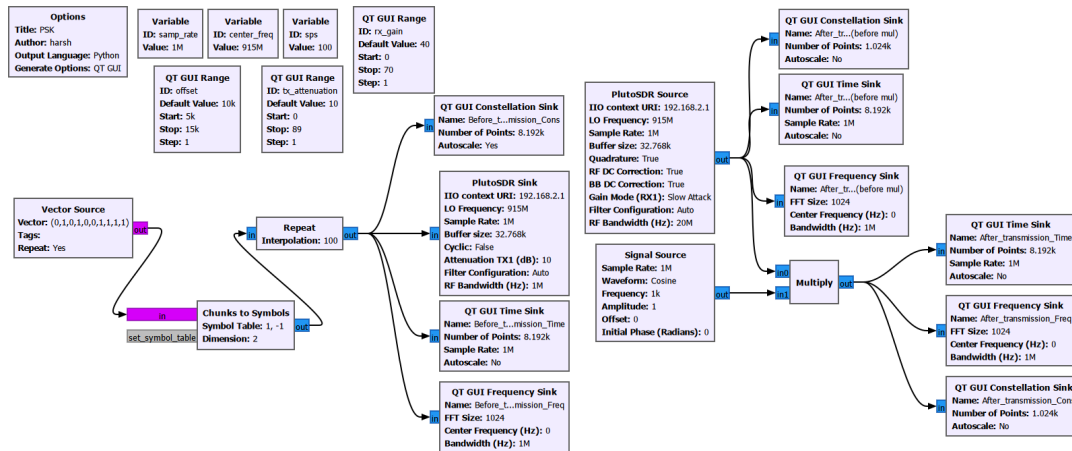


Fig. B3: GNU Radio Companion flow-graph for PSK transmitter and receiver.

5. Procedure

- The ADALM-Pluto SDR was connected to the PC via USB and detected in GNU Radio Companion (IIO context identified).
- A GRC flow-graph was created for the transmitter: a bit source was generated and mapped to the required OOK / ASK / PSK symbol levels and fed to the Pluto SDR Sink block, with an appropriate carrier/centre frequency, sample rate, and gain configured.
- A corresponding receiver flow-graph was created using the Pluto SDR Source block, followed by an AGC block, a Constellation Decoder, and slicing/threshold logic to recover the transmitted bits.
- QT GUI Time Sink, Frequency Sink and Constellation Sink blocks were added at appropriate points (before and after transmission) to visualise the signal.
- The transmitter and receiver flow-graphs were executed simultaneously, and the RF signal was transmitted and received over the air using the Pluto SDR's TX and RX ports/antennas.
- The above steps were repeated for each of the three modulation schemes — OOK, ASK, and PSK — one at a time.
- Time-domain waveforms, frequency-domain spectra, and constellation diagrams were captured before and after transmission for each scheme, and compared.

6. Observations

6.1 ASK — Observed Waveforms

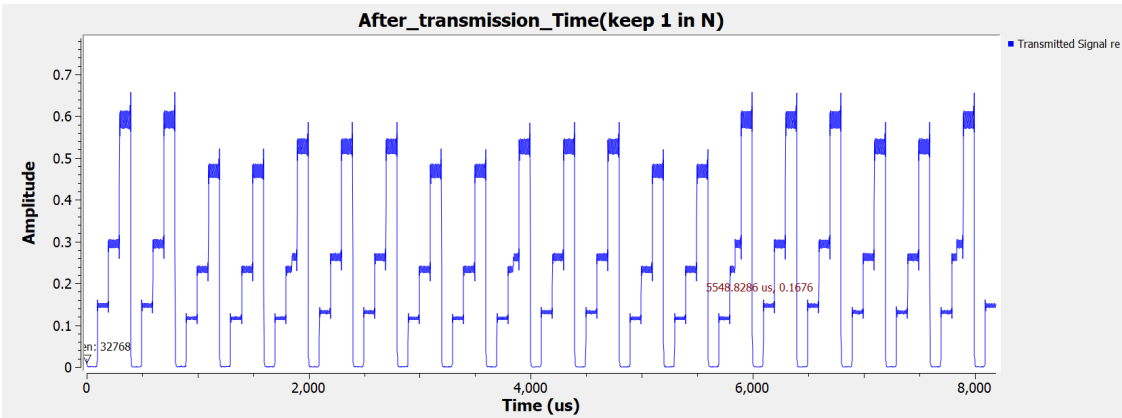


Fig. 1: Received signal after 'Keep 1 in N' block (down-sampled time-domain output) — ASK.

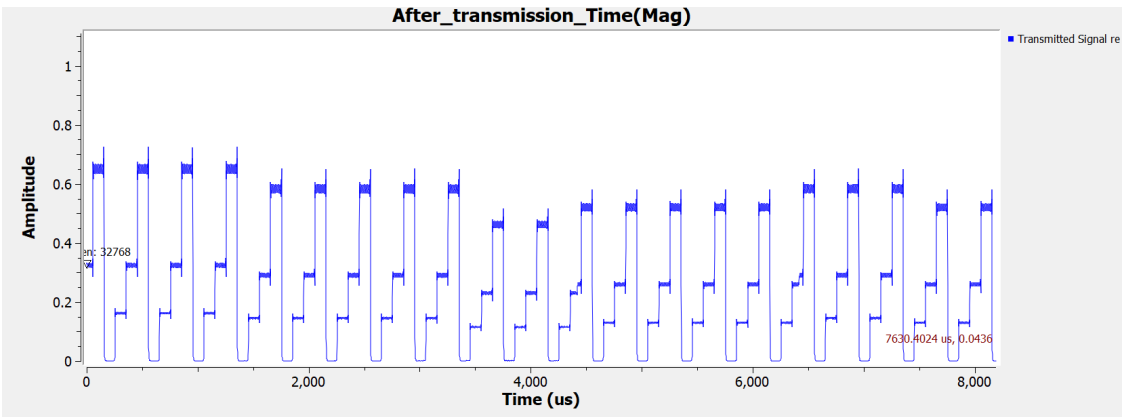


Fig. 2: Magnitude of the received signal in the time domain (After transmission) — ASK.

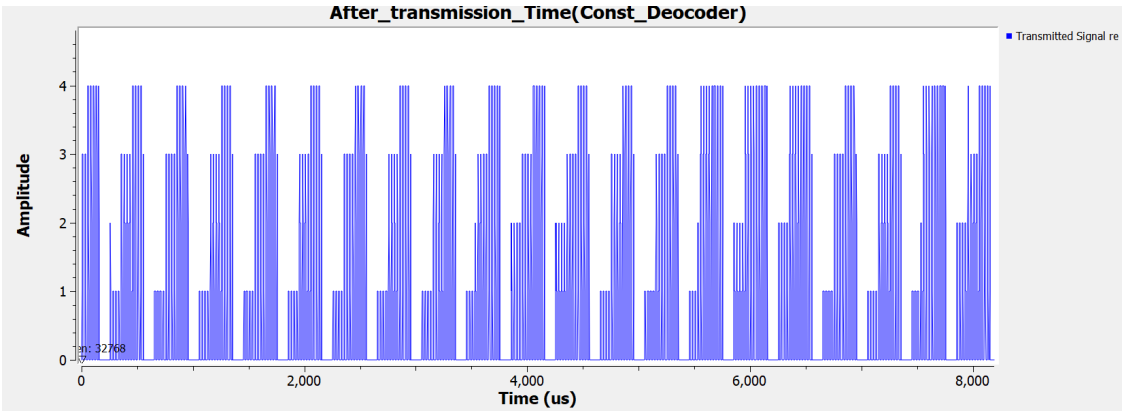


Fig. 3: Output of the Constellation Decoder block plotted against time — ASK.

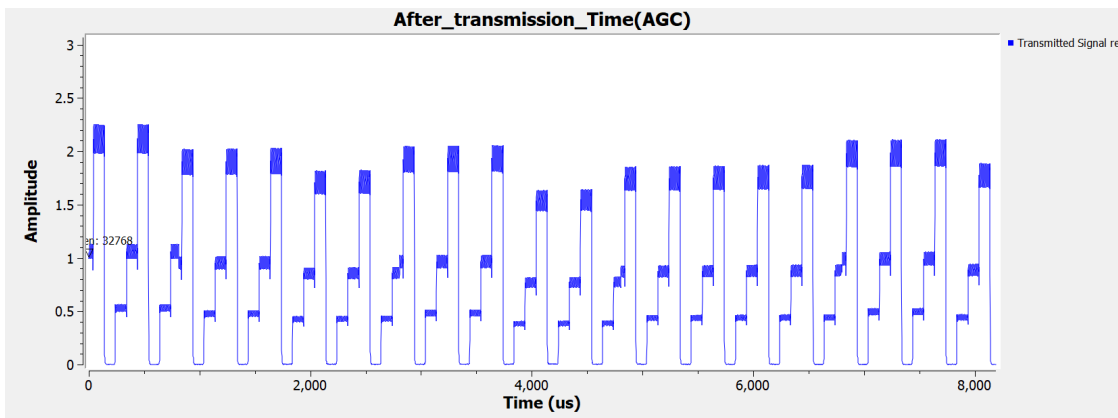


Fig. 4: Output of the AGC block (After transmission, time domain) — ASK.

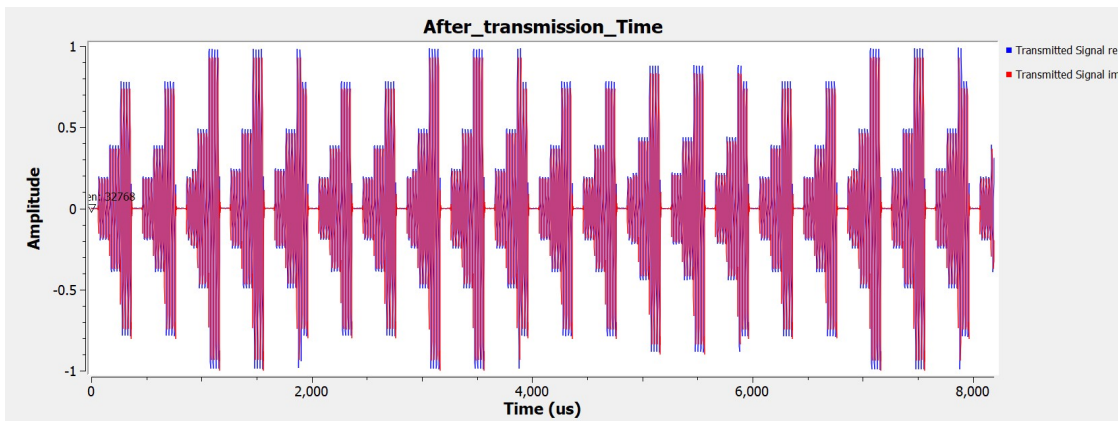


Fig. 5: Time-domain I (real) and Q (imaginary) components of the transmitted signal after passing through the Pluto SDR — ASK.

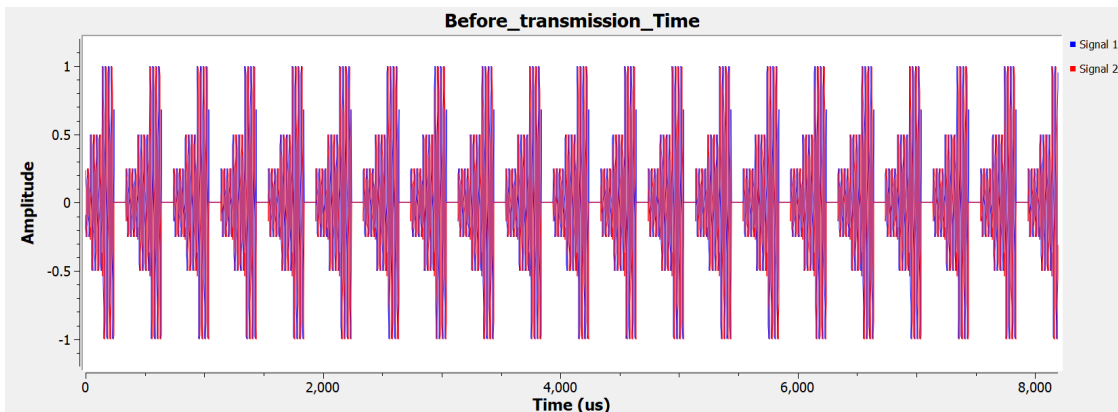


Fig. 6: Time-domain I/Q waveform of the signal before transmission (baseband) — ASK.

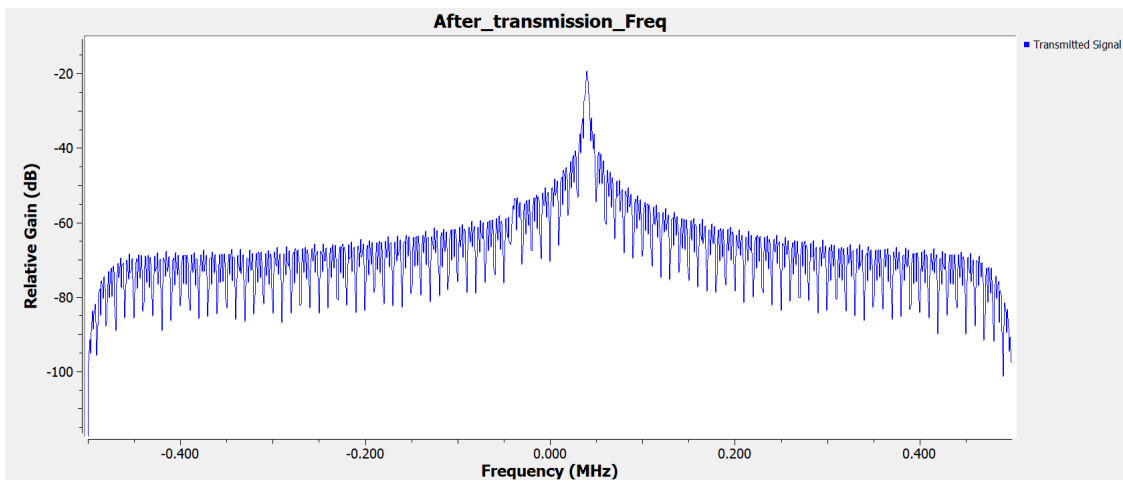


Fig. 7: Frequency spectrum of the signal after transmission — ASK.

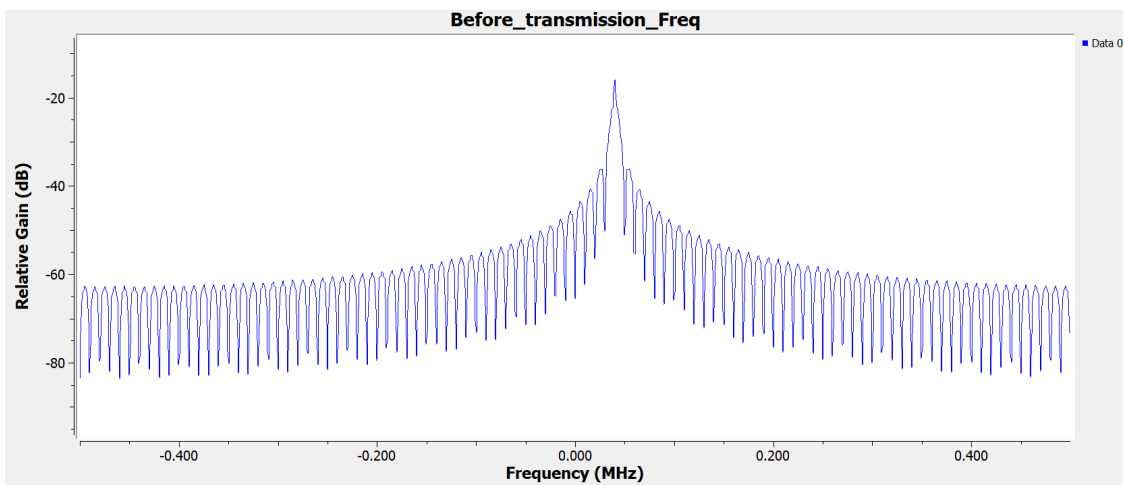


Fig. 8: Frequency spectrum of the signal before transmission (baseband) — ASK.

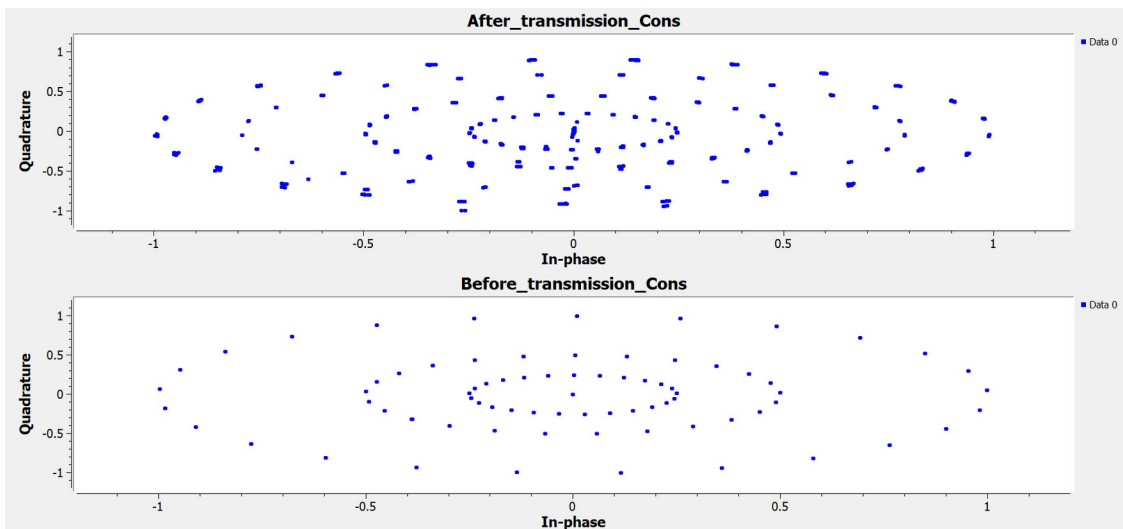


Fig. 9: Constellation diagrams before and after transmission — ASK.

6.2 OOK — Observed Waveforms

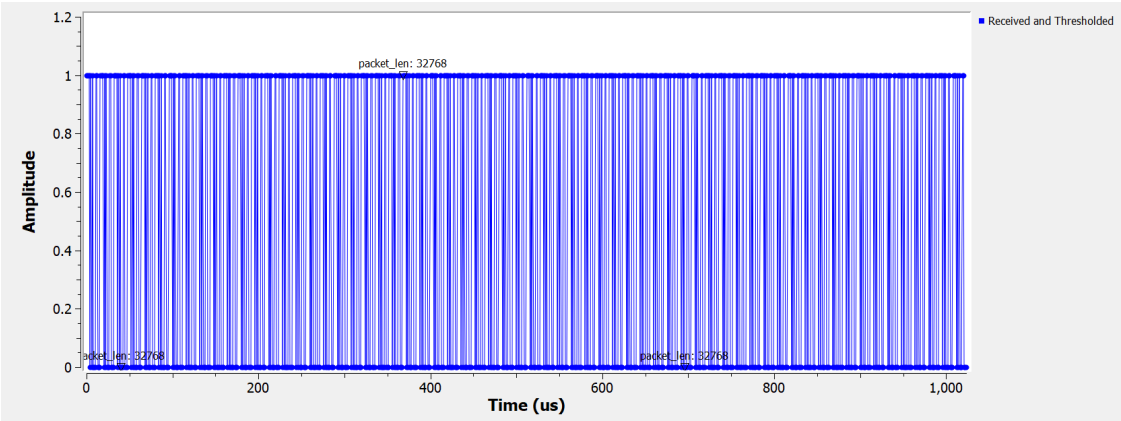


Fig. 10: Received and thresholded bit stream recovered at the receiver — OOK.

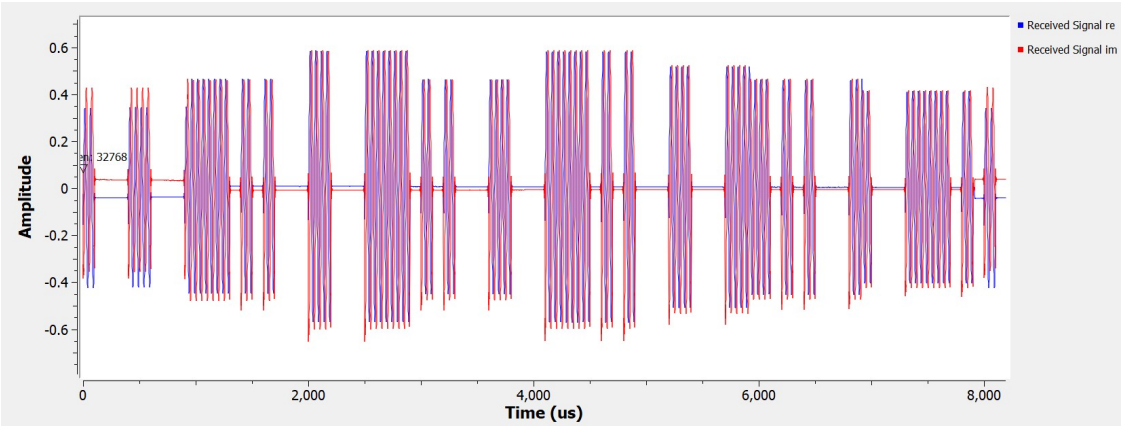


Fig. 11: I (real) and Q (imaginary) components of the received signal — OOK.

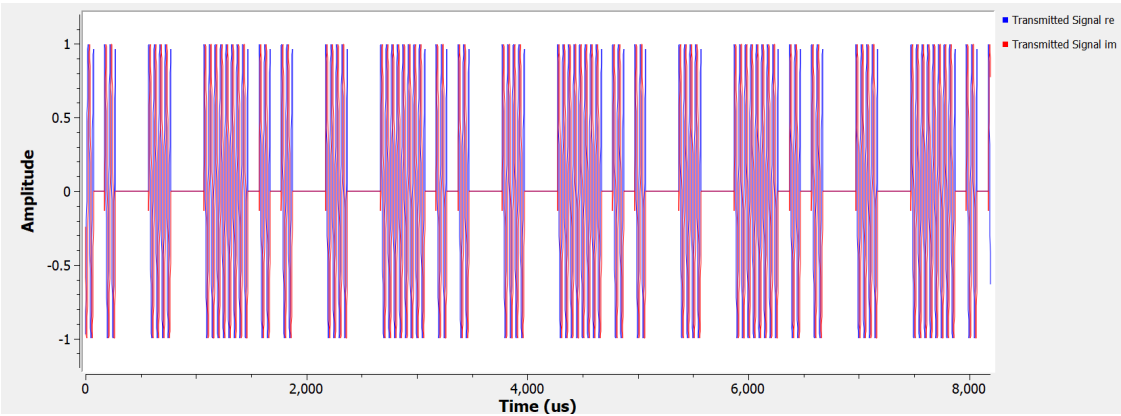


Fig. 12: I (real) and Q (imaginary) components of the transmitted signal — OOK.

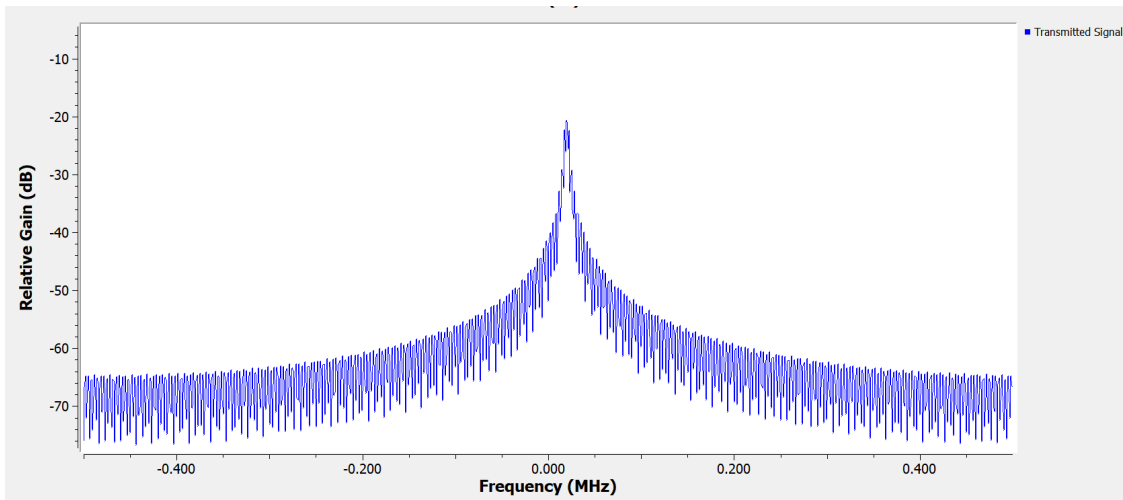


Fig. 13: Frequency spectrum of the transmitted signal — OOK.

6.3 PSK — Observed Waveforms

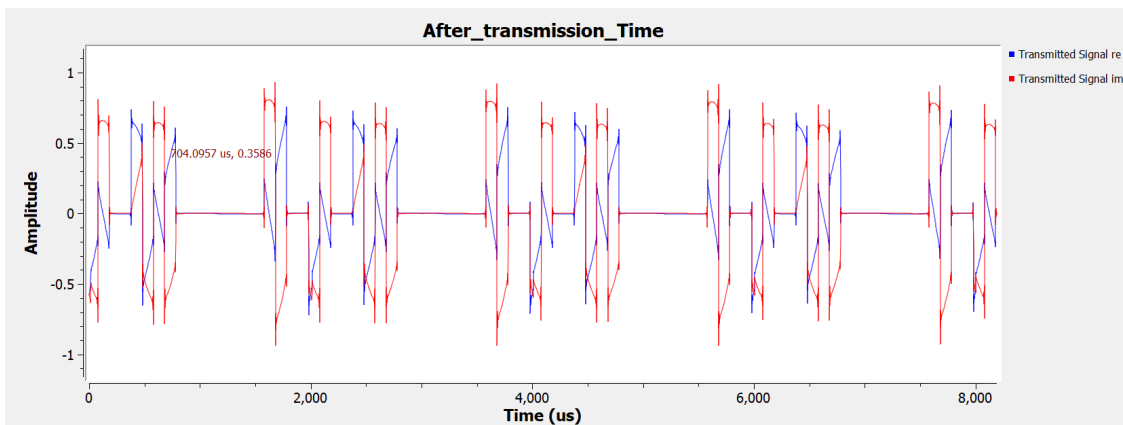


Fig. 14: Time-domain I/Q waveform of the received signal after the multiplier (After transmission) — PSK.

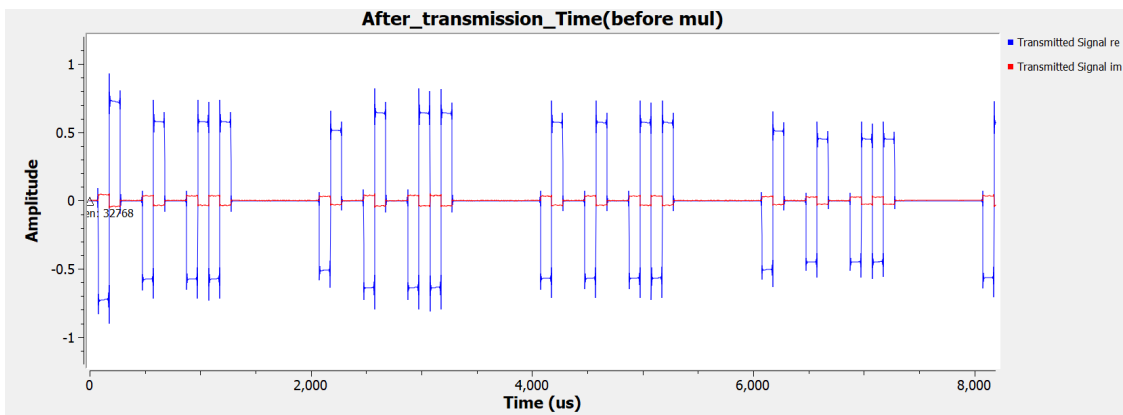


Fig. 15: Time-domain I/Q waveform of the received signal before the multiplier (After transmission) — PSK.

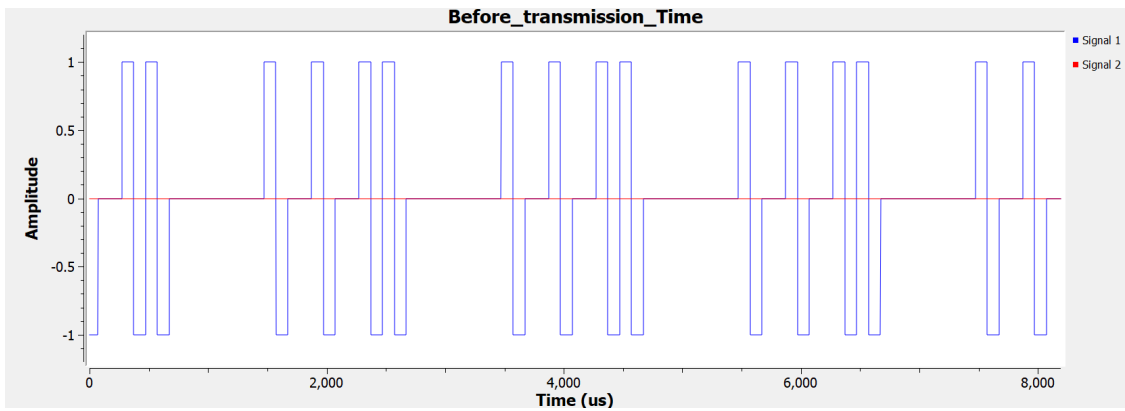


Fig. 16: Time-domain baseband bit stream before transmission — PSK.

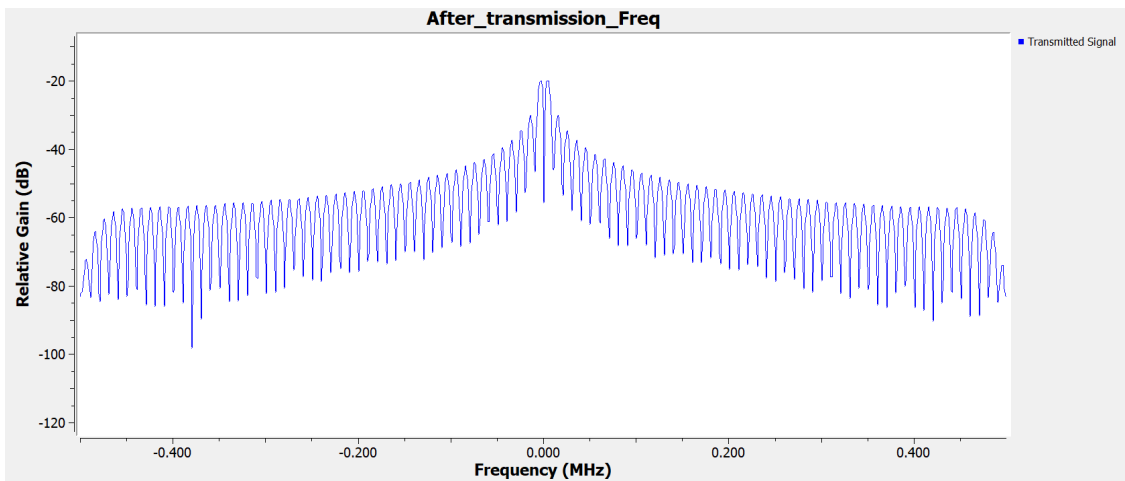


Fig. 17: Frequency spectrum of the received signal after the multiplier (After transmission) — PSK.

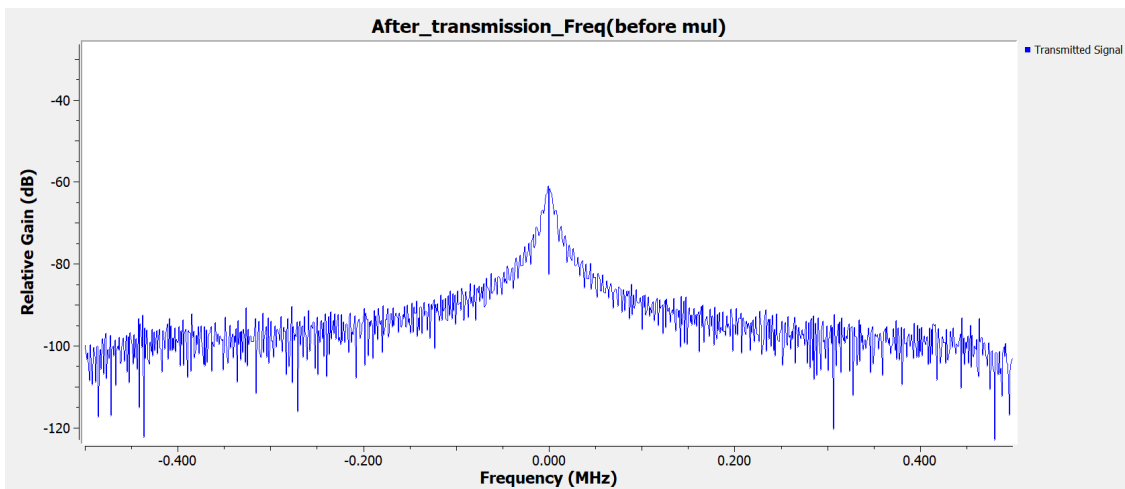


Fig. 18: Frequency spectrum of the received signal before the multiplier (After transmission) — PSK.

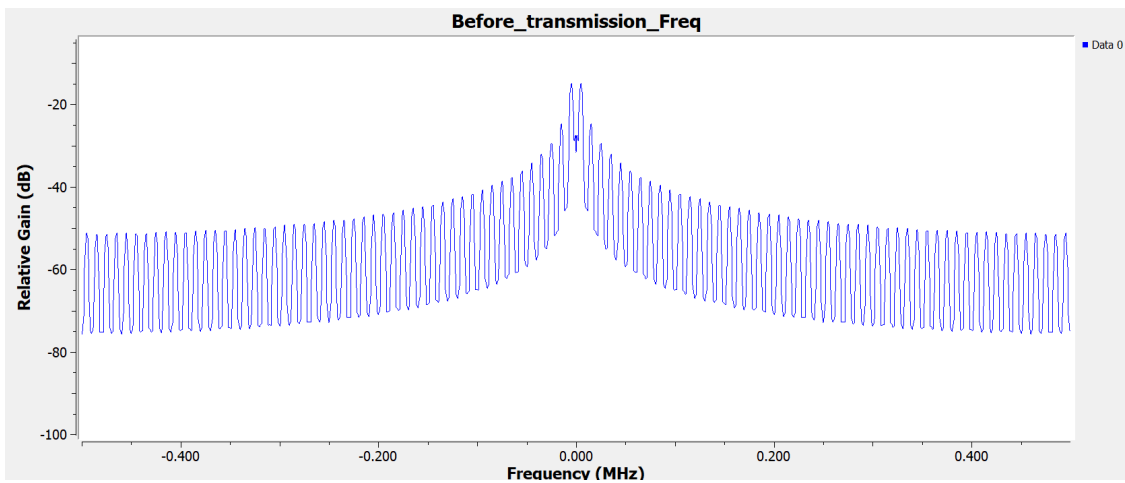


Fig. 19: Frequency spectrum of the signal before transmission (baseband) — PSK.

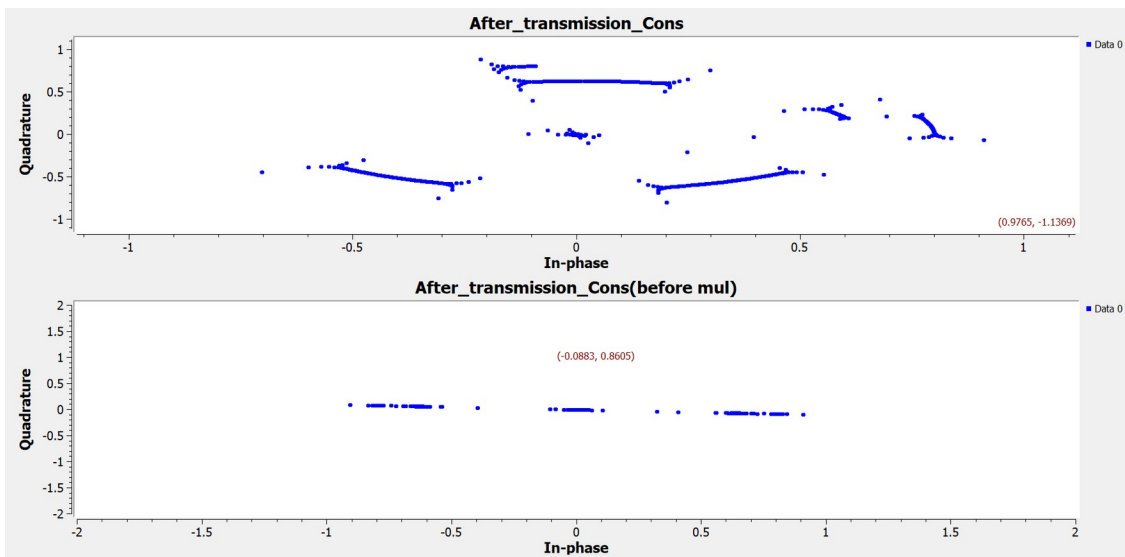


Fig. 20: Constellation diagrams after transmission — after the multiplier (top) and before the multiplier (bottom) — PSK.

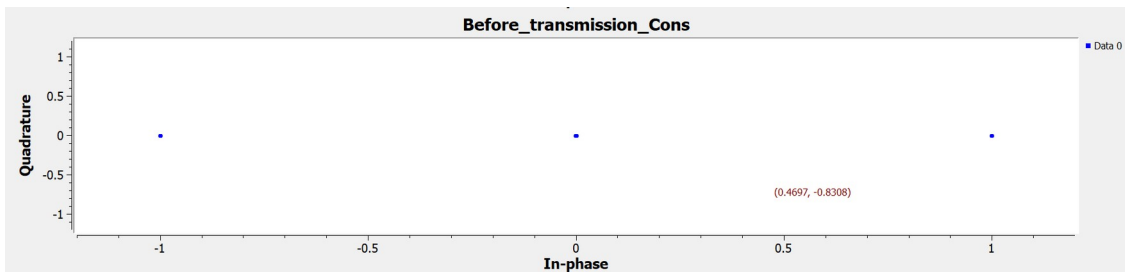


Fig. 21: Constellation diagram before transmission (ideal BPSK symbol points) — PSK.

7. Results & Discussion

For all three schemes, the transmitter flow-graph generated the required baseband signal, which was successfully transmitted over the air using the Pluto SDR Sink and recovered using the Pluto SDR Source at the receiver, confirming a working real-time RF link.

From the ASK time-domain plots (Fig. 5 and Fig. 6), it is observed that the transmitted signal amplitude fluctuates between multiple discrete non-zero levels, which is characteristic of amplitude shift keying, while the AGC output (Fig. 4) shows that the receiver automatically normalises the gain of the incoming signal to maintain a stable amplitude before demodulation. The 'Keep 1 in N' and magnitude plots (Fig. 1, Fig. 2) confirm that the down-sampled envelope of the received signal correctly follows the transmitted amplitude pattern.

For OOK, the received and thresholded plot (Fig. 10) clearly shows the recovered binary bit stream toggling between logic '0' and logic '1' levels, verifying successful envelope-based detection at the receiver. The transmitted and received I/Q time-domain plots (Fig. 11, Fig. 12) show that the carrier is completely switched off for the '0' symbol and switched on at full amplitude for the '1' symbol, which is the defining characteristic of on-off keying.

Comparing the frequency-domain spectra (Fig. 7, Fig. 8, Fig. 13), a distinct peak is observed at the carrier/centre frequency in all cases, with side-lobes spreading symmetrically on either side due to the finite symbol rate and pulse shaping used. The spectral shape after transmission shows a slightly raised noise floor compared to the ideal baseband spectrum, which can be attributed to the RF front-end noise, the wireless channel, and hardware impairments of the Pluto SDR.

The constellation diagrams (Fig. 9) provide the clearest basis for comparing the three schemes. Before transmission, the ideal ASK/OOK constellation points lie on the real (in-phase) axis at fixed, well separated amplitude levels. After transmission, the received constellation points are seen to spread out into small clusters around the ideal points instead of being exact single points; this spreading is caused by channel noise, carrier phase/frequency offset, and imperfect synchronisation introduced by the real RF link. Nevertheless, the clusters remain distinguishable, which is why the transmitted bits could still be correctly recovered at the receiver using the Constellation Decoder and threshold/slicer blocks.

For PSK, the ideal baseband constellation (Fig. 21) shows three well-separated points on the in-phase axis at -1, 0 and +1 (corresponding to the Chunks-to-Symbols mapping used, i.e. logic levels mapped to ± 1 with an unmodulated/reference symbol at 0), confirming that phase information for BPSK is encoded purely along the in-phase axis with no quadrature component before transmission. After transmission (Fig. 20), the received constellation — both before and after the multiplier stage — no longer collapses to three tight points; instead the samples trace curved, smeared arcs and clusters spread across the I-Q plane. This spreading is caused by the residual carrier frequency/phase offset between the Pluto SDR transmitter and receiver local oscillators, which was not fully compensated by a carrier recovery loop in this flow-graph, in addition to channel noise. The time-domain plots (Fig. 14–16) confirm that the modulated bursts are still recovered with the correct envelope/timing, and the frequency-domain plots (Fig. 17–19) continue to show a clear carrier peak with the expected sinc-shaped side-lobes, indicating that the RF link itself was working correctly even though the phase reference drifted.

This highlights a key practical requirement of PSK systems: because information is carried in absolute phase, a coherent receiver needs an accurate carrier phase/frequency recovery loop (e.g., a Costas loop) to 'undo' the LO phase/frequency offset before the constellation can be correctly sliced. In this experiment, without such a recovery loop, the ASK and OOK receivers (which rely on amplitude/envelope information) show more stable constellations and bit recovery than the PSK receiver (which relies on absolute phase). This is consistent with a rotated/smeared BPSK constellation being expected

whenever there is an uncompensated carrier offset, and it explains why the received PSK constellation forms rotating arcs rather than two fixed clusters.

Overall, OOK is the simplest to implement but the least noise-robust because the '0' symbol carries no energy; ASK improves on this slightly by keeping both symbols at non-zero amplitude, but is still an amplitude-based scheme and therefore sensitive to channel gain variations; PSK, by encoding information in phase instead of amplitude, is theoretically the most noise-immune of the three, but in this implementation it also proved to be the most sensitive to carrier phase/frequency offset between the transmitter and receiver, which visibly distorted its received constellation in the absence of a dedicated carrier-recovery block.

8. Conclusion

OOK, ASK, and PSK digital modulation and demodulation schemes were successfully implemented using GNU Radio Companion block diagrams and were demonstrated over a real-time RF link using the ADALM-Pluto SDR. The transmitted and received signals were captured and analysed in both the time and frequency domains, and the constellation diagrams before and after transmission were compared for each scheme. For OOK and ASK, the received constellations and thresholded bit streams showed noise-induced spreading around the ideal symbol points while still remaining separable enough for correct bit recovery using simple amplitude-based detection. For PSK, the received constellation showed noticeable smearing and rotation relative to the ideal three-point baseband constellation, caused mainly by residual carrier phase/frequency offset between the transmitter and receiver, underlining the practical need for a carrier-recovery (e.g., Costas loop) stage in a coherent PSK receiver. Overall, the experiment confirmed that while PSK is theoretically the most noise-immune scheme of the three since it does not rely on amplitude information, it is also the most sensitive to carrier synchronisation errors, whereas OOK and ASK, being amplitude-based, were comparatively easier to demodulate reliably with the simple envelope/AGC/threshold based receiver chains used here. This experiment provided practical, hands-on experience in designing SDR-based digital communication systems using GNU Radio and validating them using real over-the-air transmission with the ADALM-Pluto SDR.